

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 26)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0	1
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A motor-racing circuit has residents living 500 m away. The circuit operator wants to compare two designs of acoustic barrier: an earth baffle mound and a timber acoustic fence.

0	1
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1 Describe how you would carry out an investigation to compare the effectiveness of the two barriers.

[5 marks]

0	1
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2 Explain why readings should be taken with the same wind direction.

[1 mark]

0	1
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3 Explain the difference between the LA10 and LA90 measures of traffic noise.

[2 marks]

0	1
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4 Outline **two** ways in which the residents' houses could be designed to reduce the noise reaching occupants.

[2 marks]

1

2

10

0 2 . 1 Which of the following best describes a **representative** sample? Tick one box. [1 mark]

- A A sample taken from the most convenient location ☐
- B A sub-sample that accurately reflects the complete data set ☐
- C The largest sample that can be collected ☐
- D A sample that supports the hypothesis ☐

0 2 . 2 Describe how random sampling sites would be selected in an irregularly shaped meadow. [2 marks]

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0 2 . 3 Explain why deliberately choosing sampling sites introduces bias. [2 marks]

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5

03 . 1

Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen

☐

B Oxygen

☐

C Argon

☐

D Carbon dioxide

☐

03 . 2

Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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03 . 3

Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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5

04 . 1

Explain how a ground-source heat pump can deliver more heat energy to a building than the electrical energy used to run it.

[3 marks]

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04 . 2

Explain how Concentrating Solar Power (CSP) with thermal storage overcomes the intermittency of solar power.

[3 marks]

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04 . 3

Explain why CSP can only be used in areas with little cloud, whereas PV can be used anywhere.

[2 marks]

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04 . 4

Describe **two** environmental impacts of solar PV.

[2 marks]

1

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2

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10

0 5

Table 1 shows four general approaches to pollution control.

0 5**1**

Complete Table 1 by giving one example of each approach.

[4 marks]**Table 1**

Approach	Example
Production prevention	
Prevention of release	
Post-release remediation	
Alternative processes	

0 5**2**

Explain why prevention of production is generally preferable to post-release remediation.

[2 marks]**0 5****3**

Describe how each of the following new technologies helps to control pollution.

[4 marks]

Satellite monitoring of oil spills

GPS ship tracking

Polymer adsorption of heavy metals

Phytoremediation

10

06 . 1

Explain why abstraction from an aquifer containing 'fossil' (palaeo) water, such as the High Plains / Ogallala Aquifer in the USA, cannot be sustainable.

[3 marks]

06 . 2

Describe how the GRACE satellites are used to monitor aquifer depletion and give **one** advantage over monitoring boreholes.

[3 marks]

06 . 3

Describe **three** measures being introduced on the North China Plain to reduce aquifer depletion.

[3 marks]

1

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2

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3

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06 . 4

Suggest why large-scale afforestation on the North China Plain may reduce aquifer recharge even though trees reduce soil erosion.

[1 mark]

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10

07

Table 1 shows the per capita energy consumption and GDP per capita of five countries.

Table 1

Country	GDP per capita / US\$ thousand	Energy use per capita / GJ per year	Main energy resource
A	62	290	Oil, gas
B	44	125	Gas, nuclear, renewables
C	10	95	Coal
D	2	25	Biomass (wood), oil
E	55	220	Hydroelectricity, oil

07 . 1

Describe the relationship between GDP per capita and energy use per capita shown in Table 1.

[2 marks]

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07 . 2

Explain why affluence increases per capita energy consumption.

[3 marks]

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07 . 3

Suggest **two** reasons why country B uses much less energy per capita than country A despite a similar level of affluence.

[2 marks]

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2

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07 . 4

Explain why the per capita energy consumption of the UK has fallen in recent years even though consumption of goods has risen.

[2 marks]

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07 . 5

Give **one** way energy is used in each of the following: water supplies; agriculture.

[1 mark]

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10

08 . 1

Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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08 . 2

Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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08 . 3

Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

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08 . 4

Give **three** other positive feedback mechanisms.

[3 marks]

1

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2

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3

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08 . 5

Explain what is meant by a climate **tipping point**.

[2 marks]

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15

09 . 1

Describe **four** factors that should be considered when choosing a site for a dam and reservoir.

[4 marks]

1

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2

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3

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09 . 2

Give **one** advantage and **two** disadvantages of an estuary barrage compared with a reservoir on land.

[3 marks]

Advantage

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Disadvantage 1

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Disadvantage 2

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09 . 3

Explain why desalination of seawater is only used in some countries.

[2 marks]

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09 . 4

Explain how artificial recharge can make aquifer supplies sustainable.

[1 mark]

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10

10 . 1

Discuss how an understanding of the properties of pollutants can be used to predict their behaviour in the environment and to plan strategies to control them.

[9 marks]

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Do not write outside the box

Extra space

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10 . **2** State what is meant by **bioaccumulation**.

[1 mark]

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Write an essay on one of the following topics.

$$\boxed{1}\boxed{1}.\boxed{1}$$

Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

END OF QUESTIONS